DevDesk — Project Report for Faculty Evaluation 🖥️

Student Name: Mayank Sharma

Project Name: DevDesk (Developer Command Center)

Platform: macOS Desktop App (Apple Silicon & Intel)

Tech Stack: Electron + Next.js 16 + React 19 + Node.js/Express + SQLite

GitHub Repository: https://github.com/MAYANKSHARMA01010/DevDesk

Mac DMG Download Link: https://drive.google.com/drive/folders/1raRV1BxgL3II7Tz-kfIxTMIECMHIPDvK?usp=sharing

1. What is DevDesk? (Simple Overview)

DevDesk is a desktop application built for software developers so they don't have to keep 5 to 6 heavy applications open at the same time.

Normally, when working on a project, a developer has to open:

- **VS Code** (for writing code)
- **Terminal / iTerm** (for running `npm run dev`, `build`, etc.)
- **Postman** (for testing APIs)
- **Git GUI / GitHub Desktop** (for checking branches, commits, and diffs)
- **DBeaver or pgAdmin** (for inspecting database tables)
- **Activity Monitor / Terminal** (for finding and killing ports like 3000 or 5000 that get stuck)

Core Idea: DevDesk brings all these daily tools together into **one single desktop window**. It runs completely on the user's laptop (100% offline, local-first) with zero cloud dependency and zero tracking.

2. Why I Built This? (Problem Statement)

- **Laptop Lag & High RAM Usage:** Running VS Code + Postman + DBeaver + Terminal + Chrome eats 4 GB to 6 GB of RAM easily, making laptops heat up and slow down. DevDesk runs as a single lightweight client consuming around 350 MB.
- **Too Much Tab Switching:** Developers lose focus constantly switching between 6 different windows just to run a script, test an endpoint, or check a git diff.
- **Stuck Ports & Zombie Servers:** Often servers crash in the background and hold port 3000 or 5000. DevDesk automatically detects which ports are occupied and lets you stop them in one click.
- **Data Privacy:** Online API testers or cloud tools often sync private keys or queries to external servers. DevDesk keeps everything strictly on the local machine (`127.0.0.1`).

3. Tech Stack Used

- **Desktop Shell:** Electron Mac desktop app wrapper with native OS integration.
- **Frontend UI:** Next.js 16 React 19 Tailwind CSS (Midnight Slate dark theme).
- **Backend:** Node.js Express (runs locally on port `5001` or fallback port).
- **Local Storage:** SQLite (stores saved workspaces and project history locally in `~/Library/Application Support/DevDesk/devdesk.db`).
- **Package Manager:** pnpm Monorepo workspace configuration.

4. Key Features & How They Work (Point-by-Point)

1. Workspace & Project Scanner

What it does: You select any folder on your Mac (like `Projects` or `Downloads`), and DevDesk scans all projects inside it automatically.
How it works: Uses a recursive directory scanner (up to 4 folders deep), finds `package.json` files, identifies project types (React, Next.js, Express, Python, etc.), and lets you open any project directly in VS Code or Finder with 1 click.

2. Script Runner & Live Terminal

What it does: Reads your project scripts (like `dev`, `build`, `start`, `test`) and shows them in a clean horizontal bar. You click **Start**, and it runs the script with live colorized logs right inside the app.
How it works: Spawns background child processes using Node `child_process.spawn`. Keeps the last 1,000 log lines in a circular memory buffer so the app doesn't slow down, and lets you restart or stop the server cleanly.

3. Port & Process Manager (with In-App Browser)

What it does: Automatically checks which ports (like `:3000`, `:5000`) are active. You can open the server in your default browser, or view it directly inside an in-app interactive web preview.
How it works: Runs the system `lsof` command in the background to detect open TCP ports without freezing the UI. Only tracks processes started by DevDesk so system processes are not disturbed. The in-app preview has Desktop, Tablet, and Mobile viewport options.

4. Git Control & Push

What it does: Shows your current Git branch, changed files, line-by-line diffs (green for added lines, red for removed lines), lets you stage files, make commits, and push/pull from GitHub.
How it works: Runs local `git` commands (`status`, `diff`, `add`, `commit`, `push`) targeting the exact repository root. Also includes a **"Stage All & Commit"** and **"Commit & Push🚀 "** button so commits don't get blocked when 0 files are staged.

5. API Tester (Postman Alternative)

What it does: Lets you test REST APIs directly inside DevDesk. Supports `GET`, `POST`, `PUT`, `DELETE`, `PATCH` with query parameters, custom headers, and JSON body.
How it works: Sends HTTP requests using Node.js backend (avoiding browser CORS issues), formats the JSON response, shows status code (`200 OK`, `404`, etc.), response size, and time taken in milliseconds.

6. Environment (.env) Variables Editor

What it does: Opens and edits `.env`, `.env.local`, `.env.production` files safely with password/token masking (eye icon toggle) so secrets aren't exposed on screen.
How it works: Custom parser that keeps all comments and empty spacing intact. Before saving, it automatically creates a `.bak` backup file and writes atomically to prevent file corruption.

7. Database Viewer (PostgreSQL & MySQL)

What it does: Connects to your local PostgreSQL or MySQL database, shows tables, column schemas (primary keys, types), and lets you run queries to inspect data.
How it works: Connects via `pg` and `mysql2` drivers. **Safety feature:** Uses an SQL AST parser that only allows safe read-only queries (`SELECT`, `SHOW`). It automatically blocks accidental destructive queries like `DROP TABLE` or `DELETE`. Passwords are kept only in RAM for the active connection and are never saved to disk.

5. System Architecture (How it Works Internally)



6. Project Deliverables & Links

- **Mac Application Installer (DMG):** Self-contained `.dmg` installer for Apple Silicon Macs (~140 MB). Bundles all 129 production dependencies inside `Contents/Resources/backend` so it works out of the box without installing Node or running terminal commands.
- **Google Drive DMG Download Link:** https://drive.google.com/drive/folders/1raRV1BxgL3ll7Tz-kflxTMIECMHIPDvK?usp=sharing
- **GitHub Repository:** https://github.com/MAYANKSHARMA01010/DevDesk